



Micro Services- Interview Questions

MONOLITHIC APPLICATIONS

- 1] What is a monolithic application?
- 2] What are the key characteristics of monolithic architecture?
- 3] Why are monolithic applications tightly coupled?
- 4] What are the advantages of monolithic applications?
- 5] What are the disadvantages of monolithic applications?
- 6] Why is scaling difficult in monolithic systems?
- 7] What is deployment complexity in monolithic architecture?
- 8] How does module dependency work in monoliths?
- 9] Why do monolithic applications have long release cycles?
- 10] What is the single point of failure in monolithic architecture?
- 11] Why testing becomes complex in monolithic applications?
- 12] What is the difference between layered architecture and monolithic architecture?
- 13] Give two real-world examples of monolithic systems.
- 14] What is the problem with database coupling in monoliths?
- 15] Why monoliths have poor fault isolation?

MICROSERVICES

- 16] Why do companies move from monolithic to Microservices?
- 17] What problems do Microservices solve that monoliths cannot?
- 18] How do Microservices improve scalability?
- 19] How Microservices support faster deployment cycles?
- 20] Why Microservices are better for large teams?
- 21] How Microservices enable independent development?
- 22] Why Microservices support technology diversity?
- 23] Why Microservices reduce system downtime?
- 24] How Microservices help in fault isolation?
- 25] Why Microservices improve maintainability?
- 26] What business advantages do Microservices provide?
- 27] Why Microservices enable better cloud adoption?
- 28] How Microservices enhance agility?

29] Why Microservices support dev-ops culture?

30] Why Microservices are more suitable for distributed systems?

MICROSERVICES CHARACTERISTICS

31] What are the key characteristics of Microservices?

32] What does "loose coupling" mean?

33] What is "high cohesion" in Microservices?

34] Why Microservices must be independently deployable?

35] What does decentralized governance mean?

36] Explain decentralized data management.

37] What is continuous delivery in Microservices?

38] What is lightweight communication?

39] Explain service autonomy.

40] What is the principle of "smart endpoints and dumb pipes"?

41] Why Microservices need statelessness?

42] What is observability in Microservices?

43] Why is automation important in Microservices?

44] What is the concept of bounded context?

45] What is independent testability?

DEVELOPING MICROSERVICES

46] What are the steps to develop a Microservice?

47] Why is API design important?

48] What is DTO and why is it used?

49] Why Microservices use REST?

50] What is request validation?

51] How do you implement exception handling?

52] What is API versioning?

53] What is the role of service layer in Microservices?

54] How to handle pagination in APIs?

55] What is idempotency in Microservices?

56] Why are Microservices stateless?

- 57] How do you handle synchronous communication?
- 58] How do you handle asynchronous communication?
- 59] What is message serialization?
- 60] Why do Microservices need proper logging?

SPRING CLOUD

- 61] What is Spring Cloud?
- 62] Why is Spring Cloud popular for Microservices?
- 63] What problems does Spring Cloud solve?
- 64] How Spring Cloud helps in service discovery?
- 65] How Spring Cloud helps in load balancing?
- 66] Why Spring Cloud provides distributed configuration?
- 67] What is the role of Spring Cloud in communication between services?
- 68] How Spring Cloud improves resiliency?
- 69] What is the benefit of Spring Cloud with Spring Boot?
- 70] Why Spring Cloud is easier compared to pure Java solutions?

SPRING CLOUD CONFIG SERVER

- 71] What is Spring Cloud Config Server?
- 72] Why do Microservices need centralized configuration?
- 73] How does Config Server store configuration?
- 74] What is the purpose of Config Client?
- 75] Why external configurations are beneficial?
- 76] How does Config Server support profile-based configs?
- 77] How does Config Server refresh configuration dynamically?
- 78] What is the /refresh endpoint?
- 79] What is encryption & decryption in config server?
- 80] What is Git-backed configuration storage?
- 81] Why environment-specific configuration is important?
- 82] What is the bootstrap.properties file?
- 83] What happens if Config Server goes down?

84] How do you secure Config Server?

85] What are the limitations of Config Server?

SPRING CLOUD ADMIN SERVER

86] What is Spring Boot Admin Server?

87] Why is monitoring important in Microservices?

88] How Admin Server monitors health of Microservices?

89] What is the purpose of Admin Client?

90] What information is shown in Admin dashboard?

91] What is application health monitoring?

92] How to view logs in Admin Server?

93] How to view metrics using Admin Server?

94] What is the purpose of actuator?

95] How Admin Server uses actuator endpoints?

96] How do you secure Spring Boot Admin?

97] What is instance status page?

98] What is memory and thread monitoring?

99] How does Admin Server detect offline applications?

100] Why Admin Server is useful in production?

EUREKA SERVER

101] What is Eureka Server?

102] Why do Microservices need service discovery?

103] How does Eureka Server register services?

104] What is Eureka Client?

105] What is heartbeats in Eureka?

106] What is self-preservation mode?

107] How does Eureka handle service failures?

108] What is service registry?

109] What is service renewal in Eureka?

110] What happens if Eureka Server goes down?

111] What is the purpose of `eureka.client.fetchRegistry`?

- 112] What is eureka.client.registerWithEureka?
- 113] How does Eureka support load balancing?
- 114] Why Eureka is called client-side discovery?
- 115] Difference between Eureka and Consul?

SPRING CLOUD LOAD BALANCER

- 116] What is Spring Cloud Load Balancer?
- 117] Why load balancing is required in Microservices?
- 118] How does client-side load balancing work?
- 119] What is round robin algorithm?
- 120] What is weighted load balancing?
- 121] Why Ribbon is deprecated?
- 122] How Spring Cloud LoadBalancer replaces Ribbon?
- 123] What is service instance list?
- 124] How to configure load balancer properties?
- 125] What is retry mechanism in load balancing?
- 126] What is load balancing strategy?
- 127] What happens if one instance fails?
- 128] How load balancer chooses an instance?
- 129] What is caching of service instances?
- 130] What is load balancer health checking?

SPRING CLOUD GATEWAY

- 131] What is Spring Cloud Gateway?
- 132] Why do Microservices need an API Gateway?
- 133] What are the responsibilities of an API Gateway?
- 134] What is routing in Gateway?
- 135] What are Gateway Filters?
- 136] What is global filter?
- 137] What is predicate in Gateway?
- 138] What is path-based routing?
- 139] What is load-balanced routing?

- 140] What is rate limiting in Gateway?
- 141] How Gateway handles authentication?
- 142] Why Gateway is better than Zuul?
- 143] How to log requests in Gateway?
- 144] What is request transformation?
- 145] What is circuit breaker filter in Gateway?

RESILIENCE4J

- 146] What is Resilience4j?
- 147] Why Microservices need resiliency patterns?
- 148] What is circuit breaker pattern?
- 149] What is retry pattern?
- 150] What is rate limiter?
- 151] What is time limiter?
- 152] What is bulkhead pattern?
- 153] What is fallback mechanism?
- 154] How does circuit breaker prevent failures?
- 155] What is open state in circuit breaker?
- 156] What is half-open state?
- 157] What is closed state?
- 158] What is sliding window?
- 159] What is event publisher in Resilience4j?
- 160] How to configure resilience4j in Spring Boot?

MICROSERVICES SECURITY

- 161] Why security is essential in Microservices?
- 162] What is authentication vs authorization?
- 163] What is OAuth2?
- 164] What is OpenID Connect?
- 165] What is JWT?
- 166] What is access token?
- 167] What is refresh token?

- 168] What is role-based access control?
- 169] What is microservice-to-microservice authentication?
- 170] How to secure internal service communication?
- 171] What is HTTPS?
- 172] Why CORS is required?
- 173] What is API key based security?
- 174] Why session-based security is not suited for Microservices?
- 175] What is mTLS?

SPRING BOOT BATCH

- 176] What is Spring Batch?
- 177] Why is batch processing needed in Microservices?
- 178] What is job in Spring Batch?
- 179] What is step in Spring Batch?
- 180] What is item reader?
- 181] What is item processor?
- 182] What is item writer?
- 183] What is chunk processing?
- 184] What is job repository?
- 185] How Spring Batch handles failures?
- 186] What is restartable job?
- 187] What is job parameter?
- 188] What is tasklet?
- 189] What is skip policy?
- 190] What is retry policy?

APACHE KAFKA

- 191] What is Apache Kafka?
- 192] Why Kafka is used in Microservices?
- 193] What is Kafka topic?
- 194] What is Kafka partition?
- 195] What is Kafka broker?

- 196] What is producer?
- 197] What is consumer?
- 198] What is consumer group?
- 199] What is Kafka offset?
- 200] How Kafka achieves durability?
- 201] What is message retention?
- 202] What is at-least-once delivery?
- 203] What is exactly-once delivery?
- 204] What is dead letter topic?
- 205] Why Kafka is good for event-driven Microservices?

DISTRIBUTED TRANSACTIONS (SAGA)

- 206] What is a distributed transaction?
- 207] Why distributed transactions are difficult?
- 208] What is Saga pattern?
- 209] What is choreography-based saga?
- 210] What is orchestration-based saga?
- 211] What is compensating transaction?
- 212] What is eventual consistency?
- 213] Why 2-phase commit is not preferred?
- 214] What is transaction coordinator?
- 215] What is saga timeout?
- 216] How to maintain consistency across services?
- 217] What is outbox pattern?
- 218] Why dual writes are dangerous?
- 219] What is message relay?
- 220] What is idempotent consumer in SAGA?